

Auditing Frame-Level AUC in Weakly Supervised Video Anomaly Detection: Granularity, Resolution, and Scene Bias

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GitHub Code

Abstract. Frame-level area under the ROC curve (AUC) is the dominant evaluation metric for weakly supervised video anomaly detection (WSVAD). Its standard form measures whether an anomalous frame out-ranks a normal frame drawn from anywhere in the test set. We refer to this comparison as *pooled AUC*, since it aggregates frame pairs across test videos regardless of source. Pooled AUC therefore credits both event localization and differences between video sources. We audit this protocol on UCF-Crime [20] across recent state-of-the-art models spanning different backbone families. Holding each model’s frame scores fixed, we read them under three pairing granularities: global, per anomaly category, and within each video, then repeat the same three-granularity readout on zero-shot scores computed from the models’ internal representations. We assess ranking reliability with a paired video bootstrap. Three findings follow. First, pooled AUC does not reliably predict within-video anomaly localization: models with similar pooled scores exhibit large localization differences and rank reversals under stricter granularities. Second, at the benchmark’s test-split size, pooled AUC lacks the resolution to support state-of-the-art margins reported in the field. Within each backbone family, it resolves no comparison at those margins, while within-video AUC resolves several over identical predictions. Learned representations further reveal that within-video anomaly structure and detector localization are decoupled. Third, on normal footage alone, every model we examine separates videos by recording properties, such as resolution and color encoding, indicating that scene sensitivity is shared across the setting rather than specific to any architecture. We publicly release a granularity-aware protocol computable from existing predictions and scene-factor annotations for UCF-Crime.

Keywords: Video anomaly detection · Weak supervision · Evaluation metrics · Dataset bias · Representation auditing

1 Introduction

Weakly supervised video anomaly detection (WSVAD) aims to localize anomalous events in long videos while using only video-level labels during training [20,

21,23]. Progress on VAD benchmarks [20,25] is commonly summarized by a single metric: frame-level anomaly prediction, evaluated by the ROC-AUC (area under the true positive rate vs. false positive rate curve) pooled across all anomalous-normal frame pairs in the test set. AUC is threshold independent and admits a useful ranking interpretation statistically. However, its statistical consistency concerns *precision* at a given sample size, and does not guarantee that the *target objective* is being measured [8, 11, 19]. A consistent estimator may still be too imprecise to rank at a finite sample n , or can be faithful but estimating the wrong quantity.

A WSVAD model scores positively in an AUC comparison whenever an anomalous frame receives a larger anomaly score than a normal frame, regardless of whether the ordering follows the event semantics, behavior, or the footage source. This exposes two distinct failures, indistinguishable by the metric. First, a high AUC may not imply genuine anomaly detection. As the credited comparisons are overwhelmingly cross-video, a model’s predictions can be rewarded by AUC while learning to separate recordings rather than actual anomaly events. At deployment, a detector must rank anomalous intervals above normal ones from the same camera stream. Thus, a model relying on cross-video separation might produce constantly elevated scores, permanent false alarm, on anomaly-associated cameras while missing anomalies in atypical contexts. Pooled AUC cannot distinguish such failure from true localization. Second, the cross-video evaluation protocol yields an AUC measure too coarse to support model ranking reliably. As the easy cross-video pairs dominate the pool, they compress most models AUC toward a common narrow range, which may fall within the metric’s own sampling error [4, 7].

In this work, we investigate the evaluation process of WSVAD models. Using the same frame-level predictions produced by existing methods, we ask: (i) what does the standard pooled evaluation actually measure, and does it resolve models ranking reliably? (ii) Does the anomaly signal dominate in each model’s learned representation, and to what extent do the final anomaly scores depend on it? (iii) What determines anomaly score responses in entirely normal video segments? (iv) Does the pooled AUC ranking hold when comparisons are restricted to frames from the same recording? To answer these questions, we make the following contributions:

- A multi-granularity evaluation framework that reads the same predictions under different pairing rules, disentangling cross-recording based separation from within-video anomaly localization. We apply this to both final scores and training-free prototypes constructed from learned representations.
- A resolution analysis of WSVAD benchmarks: by resampling test videos, we test which pairwise model comparisons are statistically distinguishable at each granularity. We reveal that standard pooled AUC cannot separate models at the sub-point margins the field reports, whereas within-video AUC can.

- A scene-reliance audit of over thirteen annotated binary scene factors, computed on normal frames only. We show that every examined model separates frames by recording properties in the absence of anomalies.
- We publicly release (i) the scene-factor annotations for the full UCF-Crime dataset, and (ii) code for an evaluation protocol computable from existing predictions.

2 Related Work

Weakly supervised video anomaly detection. Since the introduction of the large-scale UCF-Crime benchmark [20], weakly supervised video anomaly detection has evolved rapidly. Models have progressed from basic temporal instance selection to increasingly sophisticated mechanisms, including robust normality modeling, memory-augmented representations, and multimodal vision-language alignments [1, 10, 14, 21–24]. Across all works, methods are still predominantly ranked by pooled AUC on UCF-Crime, which is the protocol under question in this study. Rather than proposing a new VAD model, we examine whether this evaluation protocol supports the temporal-localization conclusions drawn from the same models and predictions.

Evaluation protocols for video anomaly detection. Limitations of frame-level aggregation have previously been identified from several complementary perspectives. Ramachandra and Jones [15] introduced region-based and track-based detection criteria because frame-level AUC rewards a video frame whenever any score exceeds the threshold, without establishing that the anomalous object or trajectory was correctly localized. Their analysis exposes a spatial credit-assignment problem, where the rank statistic may be computed correctly while the set of credited detections does not correspond to the intended task. Liu *et al.* [9] identify three limitations of current VAD evaluation, which are the reliance on a single-perspective temporal annotations, failure to reward early anomaly detection, and the inability of existing benchmarks to reveal scene overfitting. They address these limitations with annotation-averaged AUC/AP, Latency-aware Average Precision (LaAP), and hard-normal benchmarks constructed to test whether models generalize beyond anomaly-associated scenes. Rashvand *et al.* [16] focus on the limitation of frame-level evaluation, where it treats a video as a collection of isolated frames, while an anomaly is a coherent temporal event with an onset and duration. The authors introduce an event-centric protocol based on temporal-IoU event matching and multi-threshold event-level F_1 .

AUC and VAD model comparison. AUC estimates the probability that a randomly selected anomalous segment receives a higher score than a randomly selected normal segment. Its threshold independence and relative stability are useful, however, pooled AUC can obscure the operating region and anomaly distribution of practical interest [19]. Defenses of AUC often conflate two distinct properties: (1) the *prevalence invariance*, which represents the positive-negative

ratio within the estimand [8, 11], and (2) the *sampling precision* of the estimator at finite test size n . Only the former is granted by AUC’s ranking interpretation. The latter was quantified analytically by Hanley and McNeil [7, 12], showing that the sampling standard errors scale as $\frac{1}{\sqrt{n}}$ in the effective sample size, regardless of positive-negative sample balance. The effective samples size in temporally correlated data, such as video frames, is closer to the number of videos than the number of frames. At the scale of current WSVAD test sets, the standard sampling error might be order of magnitudes larger than the margins by which SoTA is currently claimed. Our video-level bootstrap verifies this empirically.

Benchmark, scene, and representation auditing. A parallel line of work asks whether benchmark performance reflects the intended evidence rather than dataset-specific shortcuts [5, 6, 17, 26]. Mumcu *et al.* [13] question whether multi-scene anomaly recognition is appropriately formulated, arguing that scene and anomaly semantics can become entangled under current benchmarks. This concern motivates our scene-reliance audit, but our analysis is performed at the evaluation level.

3 Granularity-aware VAD Evaluation

Let $s_i \in \mathbb{R}$ be the anomaly score of frame i , $y_i \in \{0, 1\}$ its frame-level label, v_i its video identity, and c_i the anomaly category of its video. We compute AUC at three granularities by changing the anomaly score source, and the set of eligible positive-negative frame pairs.

3.1 Prediction-Score Granularities

We evaluate a model’s frame-level scores under three pairing rules for AUC that differ only in which anomalous–normal frame pairs are eligible for comparison: *global* (any two videos), *per-category* (same anomaly category), and *within-video* (same recording). For all, models scores are fixed and we compute the same ROC-AUC statistic varying only the population. We use *global* and *pooled* interchangeably for the standard protocol.

Global AUC. The standard frame-level protocol established in the WSVAD field [20], which pools the full test set frames:

$$\text{AUC}_{\text{global}} = \Pr(s^+ > s^-), \quad (1)$$

where $\Pr(\cdot)$ denotes probability under a uniform random draw of one anomalous frame score s^+ and one normal frame score s^- from the frame pool in the test set. Since the anomalous and normal frames may originate from any video; behavioral and contextual separability are both rewarded in $\text{AUC}_{\text{global}}$.

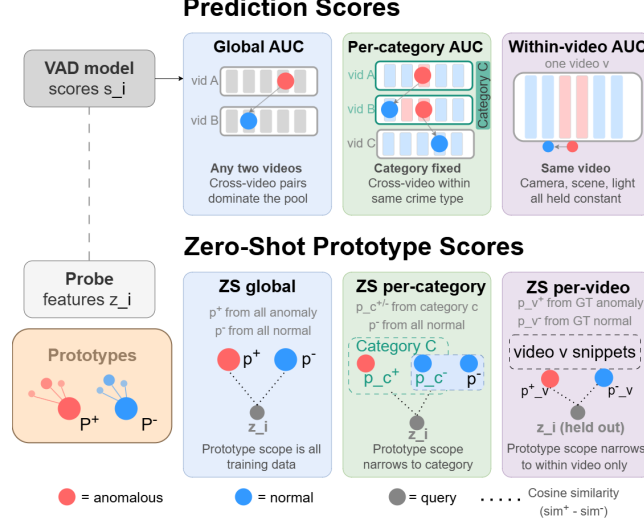


Fig. 1: Two score sources for AUC under three pairing rules. Each scoring mechanism ranks anomalous (red) against normal (blue) frames, differing only in which pairs are eligible. Top: the detector’s own prediction scores s_i . Bottom: prototype scores, where the pre-classification head embeddings from frozen VAD models are applied to construct a normal and abnormal class-mean (prototype). Then, an anomaly score for a snippet feature z_i is computed by its similarity to anomalous versus normal prototypes.

Per-category AUC. For anomaly category c , positives are the anomalous frames of videos labelled c , and negatives are the normal frames of those same videos; normal frames inherit the category of the video they occur in, and videos labelled Normal contribute nothing at this granularity. A single AUC is computed over each category’s full frame pool. The reported value averages these across categories,

$$\text{AUC}_{\text{cat}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \text{AUC}(\{s_i : y_i = 1, c_i = c\}, \{s_i : y_i = 0, c_i = c\}), \quad (2)$$

so that each category contributes equally regardless of how many videos or frames it contains. Fixing the category constrains the type of event that the positives depict, but leaves the recording context free. Specifically, an anomalous frame from one video is still ranked against a normal frame from another. This granularity is therefore stricter than global pooling with respect to category composition, while still admitting cross-video comparisons.

Within-video AUC. Let $\mathcal{V}_{\pm}$ contain test videos with at least one anomalous and one normal frame. We compute an AUC inside every video and macro-average

across videos:

$$\text{AUC}_{\text{wv}} = \frac{1}{|\mathcal{V}_{\pm}|} \sum_{v \in \mathcal{V}_{\pm}} \text{AUC}(\{s_i : y_i = 1, v_i = v\}, \{s_i : y_i = 0, v_i = v\}). \quad (3)$$

The compared frames share the same recording, camera, location, and broad acquisition conditions. The model must rank the anomalous interval above the video’s own normal footage. We therefore interpret AUC_{wv} as a *temporal localization measure*, not as a replacement for global detection performance.

3.2 Zero-Shot Representation Readout

Prediction scores characterize the final detector, but cannot identify whether a failure originates in the representation or the scoring head. We therefore repeat the same three-granularity readout using zero-shot prototype scores on the feature immediately before the model’s final classification head.

Let $z_i \in \mathbb{R}^d$ represent the feature vector for snippet i . For a given granularity level g , denote the anomalous and normal support sets by $\mathcal{S}_g^+$ and $\mathcal{S}_g^-$, respectively. Their corresponding prototypes are defined as the mean feature vectors of each class

$$p_g^+ = \frac{1}{|\mathcal{S}_g^+|} \sum_{j \in \mathcal{S}_g^+} z_j, \quad p_g^- = \frac{1}{|\mathcal{S}_g^-|} \sum_{j \in \mathcal{S}_g^-} z_j, \quad (4)$$

where $p_g^+, p_g^- \in \mathbb{R}^d$. We score query feature z_i by its difference in cosine similarity to the two prototypes:

$$q_i^{(g)} = \text{sim}_{\cos}(z_i, p_g^+) - \text{sim}_{\cos}(z_i, p_g^-), \quad \text{sim}_{\cos}(a, b) = \frac{a^\top b}{\|a\|_2 \|b\|_2}. \quad (5)$$

The support sets are constructed for each granularity g : globally, per anomaly category, or within the query video. Per-category and within-video prototypes are constructed with knowledge of ground truth snippet labels, whereas global prototype is based on video labels only. For within-video prototype, the evaluated query snippet is excluded from its support set through a leave-one-out procedure. We evaluate the resulting scores $q_i^{(g)}$ using Equations (1)–(3).

3.3 Resolving WSVAD Model Comparisons

The multi-granularity analysis asks whether pooled AUC reflects the intended within-video localization ability, both for the detector’s final scores and for prototype readouts of its learned representation. A separate question is whether the margins observed at any granularity are large enough to support the resulting model ranking. For such question, it is worth distinguishing two properties. First, AUC is a statistically well-understood metric that is relatively stable under

repeated sampling and invariant to class prevalence in its pairwise interpretation [8]. This stability concerns how *precisely* AUC estimates a *fixed* comparison population. Second, statistical stability of an estimator does not imply that a finite benchmark has the resolution to reliably distinguish models with scores differing by only fractions of a point. In most VAD benchmarks where the test set is finite in terms of unique video scenes, the variation of AUC estimates under different video resamples can be several magnitudes wider than the margins between state of the art estimates.

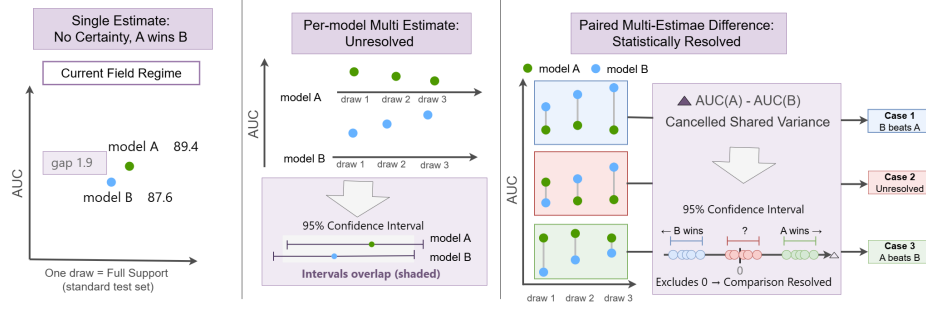


Fig. 2: AUC ranking resolution. Left: a single AUC per model. Middle: separate bootstrap sample intervals overlap, leaving comparison unresolved. Right: the difference resolves the ranking as it removes the shared effect on the video resamples.

We quantify the AUC uncertainty by repeatedly drawing evaluation videos with replacement, retaining all frames of each selected video, and recomputing AUC on each of the N resampled sets. The 2.5th and 97.5th percentiles of the resulting AUC distribution form a 95% confidence interval. Such an interval quantifies uncertainty in one model’s AUC estimate, but when intervals across models are overlapping, it does not differentiate a model from another. Therefore, model ranking requires the confidence interval of the paired AUC difference, computed from identical video resamples for both models. Figure 2 contrasts these readings on an illustrative model pair. Our paired video bootstrap and AUC difference comparison is the nonparametric counterpart to the DeLong test [4], which compares two correlated AUCs analytically via U-statistic theory, under the samples independence assumption. Since frames within a video are strongly correlated in our context, we resample at the video level to preserve this dependence structure.

Mathematically, let $\mathcal{V}$ denote the set of evaluation videos with $n = |\mathcal{V}|$. For each resample $r = 1, \dots, B$, where B is the total number of draws, we draw n videos from $\mathcal{V}$ with replacement forming the video multi-set I_r . All frames of each selected video are retained, and every model is evaluated on the same I_r . Let $\text{AUC}_g^m(I_r)$ denote the AUC of model m at granularity g computed on

replicate I_r . For a model pair (A, B) , the paired AUC difference $D_{AB}^{(r)}$ is

$$D_{AB,g}^{(r)} = \text{AUC}_g^A(I_r) - \text{AUC}_g^B(I_r). \quad (6)$$

Its percentile 95% confidence interval is obtained from the empirical quantiles ($Q_{0.025}$, $Q_{0.975}$) of these resampled differences as

$$\text{CI}_{95\%}(D_{AB,g}) = \left[Q_{0.025} \left(\{D_{AB,g}^{(r)}\}_{r=1}^B \right), Q_{0.975} \left(\{D_{AB,g}^{(r)}\}_{r=1}^B \right) \right]. \quad (7)$$

A model pair comparison is resolved at granularity g when $0 \notin \text{CI}_{95\%}(D_{AB,g})$. For model set $\mathcal{M}$, we summarize granularity g by the number of resolved unordered model pairs:

$$C_g = \sum_{\{A,B\} \subset \mathcal{M}} \mathbf{1}[0 \notin \text{CI}_{95\%}(D_{AB,g})], \quad 0 \leq C_g \leq \binom{|\mathcal{M}|}{2}. \quad (8)$$

Thus, C_g measures how many model comparisons AUC can statistically settle at granularity g .

4 WSVAD Scene Reliance Audit

The three-granularity breakdown and the AUC resolution show whether a model benefits from cross-video comparisons and whether its ranking remains consistent. However, for any resolved model comparison, none of these evaluations quantify whether the performance gain is genuine (from the use of anomaly cues), or confound (from exploiting the recording properties that correlate with the anomaly events). We tackle this by assessing model bias when it scores predefined scene factors in normal video footage.

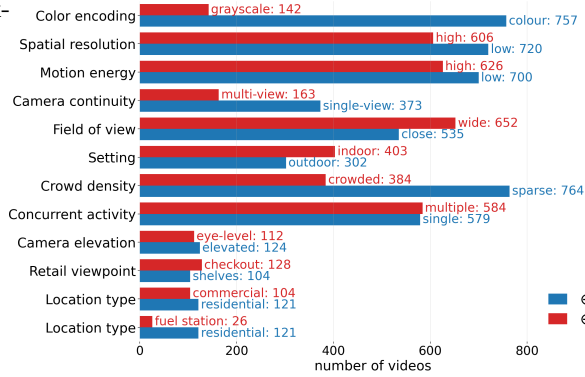


Fig. 3: Support of the scene-factor audit.

Scene factor annotation. We annotate the full UCF-Crime videos [20] with twelve binary scene factors covering acquisition conditions, viewpoint, environment, location, motion, and content composition. For each factor, a video is assigned to one of two poles ($\oplus$ or $\ominus$), or marked *ambiguous*. Ambiguous videos are excluded only from that factor, thus, each factor has its own evaluation support (Figure 3). Full annotation definitions and protocol details are provided in Supp. S1.

Table 1: Models assessed in this study. Feature backbones correspond to the original architectures adopted by the authors. AUC values are taken from the reported results, except for Universal MIL, where † indicates our implementation. Probe location is the representation immediately preceding the final linear classifier in each model.

Model	Venue	Probe location	Probe. dim	Feature backbone	AUC (%)
Universal MIL† [18]	CVPR’18	Pre-classifier hidden layer (pre- fc_2)	512	I3D	79.68
UR-DMU [23]	AAAI’23	Pre-classifier encoder ($\mathbf{x}_{\text{pre-cls}}$)	1024	I3D	86.97
BN-WVAD [24]	T-CSVT’24	Pre-classifier post-attention ($\tilde{\mathbf{x}}_{\text{att}}$)	512	I3D	87.24
PELAVAD [14]	TIP’24	Pre-classifier visual projection ($\mathbf{x}_v$)	704	I3D	86.76
π -VAD [10]	CVPR’25	Pre-classifier encoder ($\mathbf{x}_{\text{pre-cls}}$)	1024	I3D	90.33
GS-MoE [1]	ICCV’25	Pre-classifier encoder ($\mathbf{x}_{\text{pre-cls}}$)	1024	I3D	91.58
TEVAD [3]	CVPR’23	Pre-classifier fusion ($\mathbf{x}_v \oplus \mathbf{x}_t$)	2816	I3D + SwinBERT	84.92
Universal MIL† [18]	CVPR’18	Pre-classifier hidden layer (pre- fc_2)	512	VideoMAE v2 (ViT-g/14)	85.48
SST-WSVADL [2]	ECCV’26	Pre-classifier encoder ($\mathbf{x}_{\text{pre-cls}}$)	1024	VideoMAE v2 (ViT-g/14)	88.52
Universal MIL† [18]	CVPR’18	Pre-classifier hidden layer (pre- fc_2)	512	CLIP (ViT-L/14)	84.47
VadCLIP [21]	AAAI’24	Pre-classifier ($\mathbf{v} + \text{MLP}_2(\mathbf{v})$)	512	CLIP (ViT-B/16)	88.02
DSANet [22]	AAAI’26	Pre-classifier ($\mathbf{v} + \text{MLP}_2(\mathbf{v})$)	512	CLIP (ViT-B/16)	89.40
TrCLIP-VAD [18]	Neural Netw.’26	Pre-classifier ($\mathbf{f} + \text{MLP}_2(\mathbf{f})$)	1280	CLIP (ViT-L/14) + text	88.59

Measuring model bias. We compare model performance on normal frames only from different scene factor groups. Since each factor has two opposite poles, we quantify a model bias by its AUC ranking separation between two scene poles. We call this *bias-AUC*. As the anomaly label is constant across the comparison, any departure from 0.5 reflects a response to the scene factor rather than discrimination between anomalous and normal content. Writing $\bar{s}_v^0$ for the mean score over the normal frames of video v , the video-level *bias-AUC* for factor j is

$$b_j = \Pr(\bar{s}_{v_{\oplus}}^0 > \bar{s}_{v_{\ominus}}^0) + \frac{1}{2} \Pr(\bar{s}_{v_{\oplus}}^0 = \bar{s}_{v_{\ominus}}^0), \quad v_{\oplus} \sim \mathcal{V}_j^{\oplus}, \quad v_{\ominus} \sim \mathcal{V}_j^{\ominus}. \quad (9)$$

where $\mathcal{V}_j^{\oplus}$ and $\mathcal{V}_j^{\ominus}$ denote the sets of videos assigned to the two opposing poles of scene factor j . The *bias-AUC* is obtained as the ROC-AUC with pole $\oplus$ treated as the positive class. Values above 0.5 indicate higher normal-frame scores at positive pole, and below 0.5 indicates higher scores at the negative pole. We compute $d_j^{\text{bias}} = |b_j - 0.5|$ when comparing strength irrespective of sign.

Uncertainty and practical significance. We resample videos with replacement, stratified within each factor pole so that both are still represented. We keep each sampled video’s frames together to preserve temporal dependence, and recompute Equation (9) over B replicates. We report the percentile 95% interval and

$$p_{\text{bias}}^{(j)} = \frac{1}{B} \sum_{r=1}^B \mathbf{1} \left[\left| b_j^{(r)} - 0.5 \right| > 0.05 \right], \quad (10)$$

which is the fraction of replicates falling outside a practically-unbiased band of ± 0.05 . This distinguishes a factor with a small but consistent effect from one whose observed effect varies depending on which videos happened to be selected. We further comment on the effect of unequal pole sizes in Supp. S1.3.

5 Experimental Results

5.1 Dataset and Implementation Details

The headline analysis in this work emphasizes comparisons across recent top-tier WSVAD methods with different backbone families. Table 1 presents the models examined in this work. We adopt the official UCF-Crime [20] dataset for the three-granularity evaluation and scene-reliance analysis. We use the frame-level temporal annotations provided by [2]. For feature extraction, we adopt the same public I3D, VideoMAEv2, and CLIP features produced in previous works [2, 3, 10, 21, 23, 24]. Snippet size is set to 16 in all experiments. For CLIP-based variants, we substitute the center frame feature for the snippet. Global, per-category, and within-video scores are computed from the public pre-trained checkpoint predictions, except for few models remarked in Table 1, which we re-train. Prototype-based AUC refers to the AUC computed on probe features extracted from the frozen VAD models. Score-based AUC is calculated from the prediction scores of the VAD models based on the official ground truths, whereas Prob-AUC is computed on soft multi-annotator labels [9]. All AUC values are reported on the official UCF-Crime test set (290 video). The scene-factor audit in Section 4 has a larger support, including all videos from the train and test sets, to provide adequate scene diversity. The audit is descriptive rather than a performance estimate. No model is trained, selected, or calibrated on these annotations.

5.2 Anomaly Signal in Learned Representations

Table 2 (left-most column group) reports prototype-based AUC at the three granularities, computed without the model’s classification head. At the video granularity, the anomaly direction is built per video, leaving the query snippet out when constructing the normal and abnormal prototypes. Since ground truth frame labels are used to construct the prototypes, we treat AUC as a diagnostic ceiling on accessible signal, not as achievable detection performance.

The frozen backbones already contain a dominant within-video anomaly representation. VideoMAEv2 and I3D achieve a within-video ZS-AUC of 90.30 – 90.47, while CLIP reaches 88.34. Training WVAD models on these features does not consistently strengthen this signal. Compared to VideoMAEv2, within-video ZS-AUC decreases to 71.97 for UR-DMU, 74.11 for BN-WVAD, and 73.62 for SST-WSVADL. This indicates that the learned representations substantially attenuate the linearly accessible anomaly structure already present in the frozen backbone. In contrast, CLIP-based models exhibit a different behavior, improving the oracle ZS-AUC over the frozen CLIP backbone. These models retain or even strengthen a locally accessible anomaly direction, despite their lower global ZS-AUC. This suggests that the observed behavior is not a universal consequence of WSVAD training and therefore cannot yet be attributed to the choice of backbone, temporal receptive field, or architectural design.

Table 2: Prototype-based AUC vs. Score-based AUC vs. Prob-AUC at three granularities. Left: zero-shot prototype AUC on internal features. Middle: the same three granularities from each model’s final scores s . Right: the same three granularities under soft multi-annotator labels [9]. Backbone-only rows (VideoMAEv2, I3D, CLIP) have no Score-based AUC or Prob-AUC entry. Model ranks are marked as **Top-1st**, **Top-2nd**, and **Top-3rd**. [†]Indicates our re-training.

Feature / model	Prototype-based AUC			Score-based AUC			Prob-AUC		
	Global	Per-cat.	Within-video	Global	Per-cat.	Within-video	Global	Per-cat.	Within-video
VideoMAEv2	83.87	49.40	90.30	—	—	—	—	—	—
UR-DMU [†] [23]	89.58	54.59	71.97	88.08	76.50	75.83	89.51	76.80	79.40
PEL4VAD [†] [14]	49.24	49.82	48.00	87.56	73.78	76.80	89.26	73.12	76.77
BN-WVAD [†] [24]	84.75	51.29	74.11	86.76	70.74	70.44	88.76	71.13	74.10
SST-WSVADL [2]	74.72	54.56	73.62	88.52	78.11	75.75	89.48	77.88	78.67
GS-MoE [†] [1]	71.66	51.19	69.59	87.95	75.99	76.62	89.37	77.72	79.36
Pi-VAD [†] [10]	74.95	47.98	74.95	87.99	77.58	77.83	89.57	75.68	76.99
I3D	69.53	51.12	90.47	—	—	—	—	—	—
UR-DMU [23]	73.35	54.57	68.17	86.97	74.22	74.13	86.03	74.67	73.89
PEL4VAD [14]	68.27	56.42	86.94	86.76	73.22	75.42	86.87	71.54	73.07
BN-WVAD [†] [24]	73.62	55.51	71.44	84.01	65.37	72.10	85.65	66.25	70.79
SST-WSVADL [2]	75.36	60.54	68.98	86.50	74.46	73.73	85.76	74.85	73.33
GS-MoE [1]	76.02	56.35	70.65	91.58	83.86	86.79	89.26	80.25	82.59
Pi-VAD [10]	73.24	49.31	70.81	90.33	76.88	79.66	87.31	72.68	77.70
TEVAD [3]	77.47	48.02	88.79	84.92	66.53	70.79	84.61	66.25	70.79
CLIP	68.55	—	88.34	—	—	—	—	—	—
VadCLIP [21]	75.74	54.47	92.71	87.79	70.30	67.64	89.47	71.39	74.75
DSANet [22]	78.89	51.13	90.75	89.44	75.74	69.81	90.10	75.92	75.61
TrCLIP-VAD [18]	78.62	50.78	93.10	88.59	72.01	73.37	89.66	70.18	75.36

Across backbone families, per-category ZS-AUC remains close to chance, ranging from 47.98 to 60.54. This suggests that the anomaly might not be encoded as a distinct concept in the learned representations, resulting in poor transferability even across videos that belong to the same anomaly category. This holds even for models trained with explicit category-level supervision (VadCLIP and DSANet).

5.3 Representation Signal vs. Detector Performance

Section 5.2 measures the anomaly signal dominance within each model’s representations, whereas this section reads the models own anomaly scores at the same three granularities, reporting what their trained heads deliver (Table 2).

Representation signal does not predict localization. We observe that within-video score-based and prototype-based AUC disagree, and the disagreement reverses the model ranking (Table 2, *Within-video* prototype and score columns). The five models with the highest prototype AUC (86.94 – 93.10) deliver the *lowest* within-video score AUC (67.64 – 75.42). Conversely, the models with lower prototype AUC (48.00 – 74.95) deliver higher score AUC (75.75 – 86.79). The extremes make this observation concrete; PEL4VAD (VideoMAEv2) localizes at 76.80 from a prototype AUC of 48.00 (at chance), whereas VadCLIP reaches a 92.71 prototype ceiling but only 67.64 within-video score AUC. Thus, the readability of a video-specific anomaly direction from the representation appears

to be decoupled from the effectiveness of the scoring head in ranking anomalous against normal snippets in that video. This dissociation is not an artifact of the original temporal annotations. Under soft multi-annotator labels [9] (Table 2, Prob-AUC), absolute values and several ranks shift, yet the mismatch with Prototype-AUC persists. The effect is therefore robust to annotation variation on the score side.

Pooled AUC cannot separate the methods it ranks. Across all granularities in Figure S1, the per-model intervals overlap substantially. At the pooled granularity, every model’s interval contains every other model’s AUC estimate. As established in Section 3.3, overlapping intervals do not settle the models ranking. We therefore test the paired difference directly, computing Equations (6) and (8) on all models and count the number of resolved comparisons by the metric. Table 3 presents the resolution counts, and Table S4 presents detailed information on the resolved pairs.

Analyzing Tables 3 and S4 jointly, the overall resolved counts seem to overstate the AUC’s discriminative power. Among 34 resolved pairs across the model zoo, 25 are the easy cross-backbone comparisons, e.g. an I3D-based model against a CLIP-based one. Such comparisons are rarely informative, since a new method claims the state of the art by outperforming prior methods on the *same* features. Restricted to the same-backbone comparisons, global AUC resolves 0 of 15 VideoMAEv2-based pairs and 0 of 3 CLIP-based pairs, while the within-video readout resolves 5 and 1 of the same pairs. Even though the I3D family appears to be the exception (9 of 21 resolved), its resolved pairs are consistently centred on two models. The first is GS-MoE (I3D), which sits 3.5 points above the rest of the family, and the second is BN-WVAD (I3D), which sits 2.4 points further below. Among the five competitive I3D models, global AUC resolves zero pairs. Across all three families on UCF-Crime, pooled AUC seems to detect outliers separated by several points and resolves nothing at the margins the field publishes.

The stricter granularities also change which model wins. DSANet holds the highest pooled AUC of any CLIP-based detector (89.44), yet within video it is outperformed, with resolved intervals, by PEL4VAD (+7.04), GS-MoE (+6.93), UR-DMU (+6.06), SST-WSVADL (+6.04), and Pi-VAD (+7.99). None of latter models are ranked above DSANet by the pooled AUC (Table 2). The DSANet-PEL4VAD pair is unresolved at the pooled granularity and resolved within video, but in the *opposite* direction of the reported SoTA. Per-category, by contrast, largely preserves the pooled ordering among the stronger models, adding more resolutions that confirm the above pattern.

This bears directly on how improvements are claimed. DSANet is proposed as a refinement of VadCLIP and reports gains across every setting [22]. While

Table 3: Model pairs resolved under each granularity. Spread is max-min of point estimates in the subset.

Subset	Granularity	Pairs resolved	Spread (pts)
All (16 models)	Global	34 / 120	7.64
	Per-category	55 / 120	19.56
	Within-video	53 / 120	19.15
VideoMAEv2 (6)	Global	0 / 15	1.75
	Per-category	5 / 15	7.37
	Within-video	5 / 15	7.38
I3D (7)	Global	9 / 21	7.64
	Per-category	14 / 21	19.56
	Within-video	13 / 21	16.00
CLIP (3)	Global	0 / 3	1.65
	Per-category	1 / 3	5.44
	Within-video	1 / 3	5.73



Fig. 4: Scene reliance audit on normal frames only.

the two are separated per-category (+4.80, resolved), the global and within video differences are unresolved, and both models are outperformed by the same set of simpler detectors (Table S4). From observations in Table S4, no pair separated by a margin 0.4 – 0.5 AUC resolves in any backbone family, yet recent methods on UCF-Crime claim improvements within such margin [2, 14, 21, 22].

5.4 Scene Reliance of Anomaly Scores

When anomaly direction is not captured by the class-mean probe of learned representations, as shown in previous sections, the final scores may rely on anomaly-relevant evidence or on recording properties that co-vary with it. We directly

evaluate the scene bias, quantified as the deviation of the *bias-AUC* from the null band $p_{\text{bias}} \pm 0.05$ across the twelve labeled scene factors (Figure 4).

Bias to Recording Properties. Two acquisition factors are displaced in the same direction for every score sources. Colour encoding falls below 0.5 throughout (0.17 to 0.47) and spatial resolution above it (0.56 to 0.71); resolution reaches $p_{\text{bias}} = 1.000$ for every model. This unanimity spans all backbone families, architectures, and training objectives. Camera continuity, motion energy, and camera elevation follow the same pattern but at smaller magnitude. Field of view is the one acquisition factor with no bias response, where every model sits inside the practically-unbiased band (0.45–0.54).

Bias to semantic properties. Scene response is not confined to acquisition properties. Location type, crowd density, and concurrent activity displace most models above 0.55, while outdoor-indoor setting sits consistently below 0.5. These semantic and content biases are real but smaller in magnitude than the acquisition biases, and on most factors the models agree in direction. Only the location-type factors show disagreement in direction across models and backbones.

Backbone and head contributions. The bias response does not seem to originate in any trained detector. Universal MIL, a minimal head trained separately on VideoMAEv2, I3D, and CLIP features, shows similar directions on the same factors as the full models (Figure 4b). The bias magnitude shifts with the feature space more than with the model, therefore models that share a backbone cluster together. For example, the I3D columns show strong color-encoding response for GS-MoE (0.17) as well as for the minimal head (0.22). Thus, the backbone seems to set the regime, while model architectures and objectives shift values few points around it.

Impact of scene bias on anomaly detection. A scene-conditioned decomposition of AUC on anomaly-vs-normal frames shows that the measured bias rarely inflates the anomaly frames scores, except for the color encoding factor. The bias response is present in both anomaly and normal frame scores, thus, it mostly cancels in with frames pooling. See discussion in Supp. S1.2 and S2.2.

6 Conclusion

We asked what pooled frame-level AUC measures on WSVAD benchmarks, such as UCF-Crime, by reading the same anomaly scores under progressively stricter pairing rules and auditing what they respond to. Three key findings follow. First, our experiments on UCF-Crime show that the pooled AUC evaluation protocol lacks the discriminative power to support the rankings built on it. Second, the learned representation by WSVAD models and their predictions are decoupled. The models with the most dominant within-video anomaly structure may localize worst, whereas the strongest within-video detectors may produce representations that no class-mean probe can read. Pooled AUC registers neither of these

patterns. Third, every considered model separates normal footage by recording and semantic properties, such as resolution, colour encoding, and location, which carry no anomaly signal by construction.

We do not indict AUC as a statistic, since what fails mostly is the population of comparisons constructed for the pooled protocol. Dominated by frame pairs from different recordings, pooled AUC can reward separability without learning any anomaly concept. The correction is inexpensive and can be derived from the available predictions produced in previous studies. Model comparisons can be resolved by: (1) reporting within-video and per-category AUC alongside the pooled number, and (2) testing claimed improvements as paired differences on shared video resamples. We release our annotations and protocol to make this the path of least resistance.

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Storyline for *Auditing Frame-Level AUC in Weakly Supervised Video Anomaly Detection (WSVAD): Granularity, Resolution, and Scene Bias*

1. *Why interesting?*

- a WSVAD temporally localizes the frames that belong to hazardous incidents or criminal behaviors in long surveillance video.
- b In practice, decisions are made on one camera stream: an anomalous frame’s score must outrank the normal frames’ scores in that same stream.
- c Standard benchmarks, such as UCF-Crime [8], provide only video-level binary labels (anomaly/normal) over videos from diverse recordings and scenes; models trained on these labels must localize at the frame-level.
- d This encourages shortcuts: learning scene or recording properties that co-vary with anomalies rather than the anomalies themselves.
- e The evaluation protocol neither distinguishes nor flags this failure mode, yet it is still used to report new state of the art.

2. *How done now?*

- a Some work targets spatial (pixel-level) localization rather than temporal detection, but frame-level spatial annotations are scarce.
- b Other work shifts to VLM-based models that detect anomalies from contextual descriptions; the confounded protocol is still adopted, and does not separate these methods or establish significance in their reported AUC.
- c Refinements of the protocol report additional metrics alongside the main one.

3. *What is missing, and So What?*

- a Refining the protocol helps, but most reported metrics share the same problem.
- b The standard metric is pooled AUC: ROC-AUC computed over all frame-level predictions pooled across test videos; it reflects only a model’s ranking ability.
- c In a pooled comparison, the two frames of a pair come from different recordings, so pooled AUC (and any metric built on the same pairs) can be satisfied by separating recordings, not localizing events. We argue that the confound lives in the pairing, not the scoring.

4. *Proposed approach (P).*

- a Hold scores fixed; change the pairing (which frames are compared), not how they are scoring.
- b Read the same scores at three granularities: global, per-category, within-video (same recording).
- c Probe the learned representation directly from pre-trained WSVAD models, to separate what the representation encodes from what the trained head delivers.
- d Test claimed gains as paired differences on shared video resamples (video-level uncertainty).
- e Audit scene reliance directly: score normal-only frames by recording or semantic factors.

5. *Experimental questions.*

- a Does pooled AUC reward recording separation over within-video localization?
 - c1: same scores, three granularities, matched backbones.
 - c2: models rank normal-vs-anomalous well when pooled ($\approx 85-90$).
 - c3: within-video drops sharply; rankings reverse.
 - c4: learned representation probe $\perp$ delivered localization (decoupled).
- b Can the metric resolve the margins the field reports?
 - c1: paired differences on shared video resamples, per backbone family.
 - c2/c3: 0 pairs resolved at 0.4–0.5 margins; within-video resolves several.
 - c4: refining the score (Prob-AUC, soft labels) leaves ranking unchanged.
- c Does the scene confound occur in real data?
 - u1: full UCF-Crime, 12 annotated scene factors.
 - u2: reproduced baselines match published pooled AUC.
 - u3: every model separates normal-only frames by recording (resolution $p_{\text{bias}} = 1.0$).
 - u4: effect is backbone-driven, shared across architectures.

Supplementary Material

Auditing Frame-Level AUC in Weakly Supervised Video Anomaly Detection: Granularity, Resolution, and Scene Bias

No Author Given

No Institute Given

S1 Scene Reliance Audit

S1.1 Scene factor Annotation

We annotate all videos in UCF-Crime according to a set of recording properties and distinguishable scene content (Table S1). Semantic content factors are chosen empirically for UCF-Crime, based on what the dataset can support. Except for frame resolution and motion level, the rest of the factors were annotated manually. Manual annotation was performed with a purpose-built tool that plays each video and allows selection of multiple factors when present. The annotation was performed by a single annotator, where the video was watched several times at higher speeds for extra verification of the presence of a scene factor. A video is assigned a label only when the criterion holds for the majority of its duration; borderline or mixed cases are marked *ambiguous* and dropped from that factor. Ambiguous video are excluded from the concerning factor comparison but still included for others. Ambiguity was mostly present in factors, such as camera elevation, field of view, and concurrent activities.

S1.2 Scene-conditioned Anomaly AUC

As established in Section 4, *bias-AUC* indicates whether anomaly scores respond to a scene property on normal footage. However, this does not determine whether that response contributes to anomaly discrimination. For example, does a model’s anomaly identification come partly from bias to a scene factor? In other words, based on the annotated scene factors, how much of a model’s anomaly-vs-normal AUC is a scene-factor ranking? We answer this by recomputing the anomaly-vs-normal AUC with cross-pole comparisons removed, and measuring how much the AUC changes.

Setup. Fix a scene factor j with two poles $\oplus$ and $\ominus$ (e.g. indoor/outdoor), whose per-pole video supports are $\mathcal{V}_j^\oplus$ and $\mathcal{V}_j^\ominus$. Any anomalous-normal frame pair entering the pooled AUC is one of four *pair types*: (1) same-pole pairs ($\oplus\oplus$, $\ominus\ominus$), which compare frames sharing the factor; (2) cross-pole pairs ($\oplus\ominus$, $\ominus\oplus$), which compare across the factor.

Table S1: Scene factors annotated on UCF-Crime. Each factor is a binary contrast between two poles ($\oplus / \ominus$); Support is the number of videos assigned to each pole. Level 1 factors describe how the footage was recorded; Level 2 factors describe the scene content. The concurrent-activity factor is reported only as a composition control (its poles correlate with anomaly class by construction). Resolution is labelled by effective sharpness, as UCF-Crime is distributed at a uniform container resolution.

Factor	Poles	Criterion	Support
Level 1 Recording properties			
Color encoding	color / grayscale	RGB colored; IR/nighttime grayscale	757 / 142
Spatial resolution	high / low	effective sharpness: mean Laplacian variance at native resolution, median-split into high/low.	606 / 720
Camera continuity	single-view / multi-view	one continuous camera vs. cuts/-zooms/tracking	373 / 163
Camera elevation	elevated / eye-level	extremely elevated mount vs. eye-level or human-height view	124 / 112
Field of view	close / wide	tight framing (single context) vs. wide area (multiple context)	535 / 652
Motion energy	high / low	frame-difference magnitude on normal segments, terciles	626 / 700
Level 2 Content composition			
Number of people	few / many	≤ 4 vs. $5+$ visible at both normal and abnormal frames; discontinuity is allowed.	764 / 384
Setting	outdoor / indoor	exterior (street scenes) vs. private building interiors (homes, care rooms)	302 / 403
Viewpoint	shop-shelves / shop-cashair	in-store camera orientation	104 / 128
Location type [†]	residential / commercial	house facade/driveway/entrance vs. shops/business	121 / 104
Location type [†]	residential / fuel station	shared residential pole; corroboration only	121 / 26
Location type [†]	fuel station / commercial	shared poles	104 / 26
Concurrent activity	single / multiple	one vs. several simultaneous activities; must co-align with anomaly when applicable	579 / 584

[†]The two location factors share the positive pole (residential) and are not independent.

Reference AUC. The standard (pooled) AUC on factor j is a weighted average of the four pair-type AUCs,

$$AUC_{\text{ref}} = w_{\oplus\oplus} AUC_{\oplus\oplus} + w_{\ominus\ominus} AUC_{\ominus\ominus} + w_{\oplus\ominus} AUC_{\oplus\ominus} + w_{\ominus\oplus} AUC_{\ominus\oplus}. \quad (1)$$

where w is the fraction of all pooled pairs of type $(X, Y) \in \{\oplus, \ominus\}$, such that $\sum_{X,Y} w_{XY} = 1$. This is an exact identity: it regroups the same pooled AUC by pair type.

Table S2: Scene-control gap $\Delta_{\text{scene}}^{(j)} = \text{AUC}_{\text{ref}}^{(j)} - \text{AUC}_{\text{cond}}^{(j)}$ on the full prediction pool. AUC_{cond} equal-weights the two same-pole AUCs. Bold marks $|\Delta_{\text{scene}}| \geq 0.02$. Median $|\Delta|$ across all model-factor cells is 0.004. [†] Location poles are residential and commercial.

Factor	UR-DMU [12]	BN-VAD [13]	PEL [6]	SST [2]	π -VAD [5]	GS-MoE [1]	VadCLIP [10]	DSANet [11]	TEVAD [3]	TrCLIP [7]	UniMIL [8]	mean
Color encoding	+0.010	+0.028	+0.020	+0.016	+0.022	+0.021	+0.034	+0.023	+0.069	+0.036	+0.021	+0.027
Resolution	+0.000	+0.001	+0.002	+0.001	+0.002	+0.002	+0.002	+0.003	+0.004	+0.003	+0.002	+0.002
Continuity	+0.009	+0.000	+0.001	+0.007	+0.015	+0.014	+0.002	+0.007	+0.007	+0.003	+0.009	+0.007
Elevation	-0.004	+0.004	-0.003	-0.001	+0.000	+0.001	+0.007	+0.012	+0.001	+0.002	+0.012	+0.004
Field of view	+0.001	+0.002	+0.000	+0.001	-0.002	+0.000	+0.001	-0.002	-0.001	+0.002	+0.001	+0.000
Motion	+0.000	-0.001	-0.001	-0.000	+0.004	+0.002	+0.002	+0.004	+0.002	+0.003	+0.001	+0.001
Crowd	-0.009	-0.010	-0.011	-0.010	-0.011	-0.004	-0.013	-0.014	-0.014	-0.011	-0.011	-0.011
Setting	-0.002	-0.001	-0.002	-0.003	-0.002	+0.001	-0.004	-0.002	-0.003	-0.002	+0.000	-0.002
View point	-0.008	-0.011	-0.011	-0.015	-0.008	+0.002	-0.004	-0.004	-0.001	-0.005	-0.010	-0.007
#Activities	+0.015	+0.007	-0.000	+0.005	+0.007	+0.004	+0.025	+0.005	+0.026	+0.014	+0.006	+0.011
Location†	-0.010	+0.001	-0.004	-0.010	+0.002	+0.001	-0.001	+0.003	+0.002	+0.006	-0.009	-0.003

Scene-conditioned AUC. We then keep only same-pole pairs and weight the two poles equally,

$$\text{AUC}_{\text{cond}}^{(j)} = \frac{1}{2} (\text{AUC}_{\oplus\oplus} + \text{AUC}_{\ominus\ominus}), \quad (2)$$

so that every anomalous frame is ranked only against normal frames sharing its scene pole. The drop from the reference to the conditioned readout,

$$\Delta_{\text{scene}}^{(j)} = \text{AUC}_{\text{ref}}^{(j)} - \text{AUC}_{\text{cond}}^{(j)}, \quad (3)$$

measures how much of the anomaly-vs-normal separation on factor j depended on cross-pole comparisons. Since Equation (2) drops both cross-pole pairs and re-balances the two poles, $\Delta_{\text{scene}}^{(j)}$ is a protocol contrast rather than an additive decomposition of the reference AUC.

S1.3 Bias-AUC under unequal pole sizes

The scene-factor supports in Section 4 are often imbalanced, as shown in Figure 3 and Table S1. For instance, 757 colored videos vs. 142 grayscale videos. This raises a concern: could a reported *bias-AUC* far from 0.5 merely reflect this imbalance rather than a genuine scene response? We answer this briefly. *bias-AUC* estimates a pairwise ranking probability rather than a proportion. Thus, it is invariant to how many videos fall at each pole [4]. A model that ranks videos neutrally with respect to the factor obtains *bias-AUC* of 0.5 regardless of pole sizes, rather than drifting toward the majority prevalence. Imbalance instead widens the video-bootstrap interval (fewer videos in the minority pole give a noisier estimate) without shifting its expected value.

S2 Experimental Results

S2.1 Anomaly Representations in WSVAD Models

Why category-supervised models yields a per-video rather than a shared anomaly direction As reported in Section 5.2 (Tab. 2), VadCLIP and DSANet reach near-chance per-category prototype AUC despite being trained with explicit

Table S3: Mean pair-type AUCs per model, averaged over the four significant scene factors (**residential_vs_gasstation**, **color_vs_grayscale**, **singleact_vs_multiact**, **few-person_vs_crowd**; $|\text{mean } \Delta_{\text{scene}}| \geq 0.01$). Pole $\oplus$ ($\ominus$) is the first (second) group in each factor name; $\text{AUC}_{\oplus\ominus}$ ranks anomalous frames from pole $\oplus$ against normal frames from pole $\ominus$.

Model	$\text{AUC}_{\ominus\ominus}$	$\text{AUC}_{\oplus\oplus}$	$\text{AUC}_{\oplus\ominus}$	$\text{AUC}_{\ominus\oplus}$	mean Δ
UR-DMU [12]	0.826	0.779	0.801	0.776	-0.005
BN-WVAD [13]	0.807	0.770	0.749	0.819	-0.011
PEL4VAD [6]	0.815	0.791	0.776	0.837	-0.007
SST-WSVADL [2]	0.811	0.768	0.786	0.778	-0.009
π -VAD [5]	0.779	0.731	0.755	0.751	-0.005
GS-MoE [1]	0.813	0.805	0.807	0.815	+0.003
VadCLIP [10]	0.762	0.746	0.750	0.745	+0.004
DSANet [11]	0.792	0.770	0.763	0.793	-0.003
TEVAD [3]	0.741	0.697	0.762	0.679	+0.003
TrCLIP-VAD [7]	0.786	0.746	0.771	0.766	-0.005
Universal MIL [8]	0.785	0.751	0.756	0.779	-0.002

category supervision. We explain here that this follows from how that supervision is implemented rather than from a category-level anomaly direction being unlearnable. This outcome is consistent with a design choice shared by both architectures [10, 11]. Neither model scores frames against the class embeddings directly. Instead, both models first inject a visual prompt aggregated from the video’s own anomaly-weighted features into each class embedding. Thus, the direction M against which frame i is scored for category c is video-conditioned,

$$M_{i,c} = \langle x_i, t_c(v) \rangle, \quad t_c(v) = t_c + g(\text{Norm}(A^\top X)), \quad (4)$$

where X and A are video v ’s frame features and anomaly scores. The category loss is minimised whenever $t_c(v)$ separates the frames of its own video; it never requires $t_c(v) \approx t_c(v')$ across videos of the same category. Thus, the shared direction is not enforced, while the per-category prototype readout measures precisely the shared component. The local-but-not-transferable pattern thus follows from how category supervision is implemented

S2.2 Scene-Reliance Audit

Scene bias and anomaly discrimination. Table S2 reports $\Delta_{\text{scene}}^{(j)}$ for every model-factor cell. The gaps are small for all models ($|\Delta_{\text{scene}}| \leq 0.01$). Removing cross-pole comparisons barely changes the anomaly-vs-normal AUC. Therefore, the strong scene responses measured on normal footage (Section 5.4 and Fig. 4) might not translate into inflated detection scores. The bias is likely shifting anomalous and normal frames of a pole together, and canceling once frames are compared within the pole.

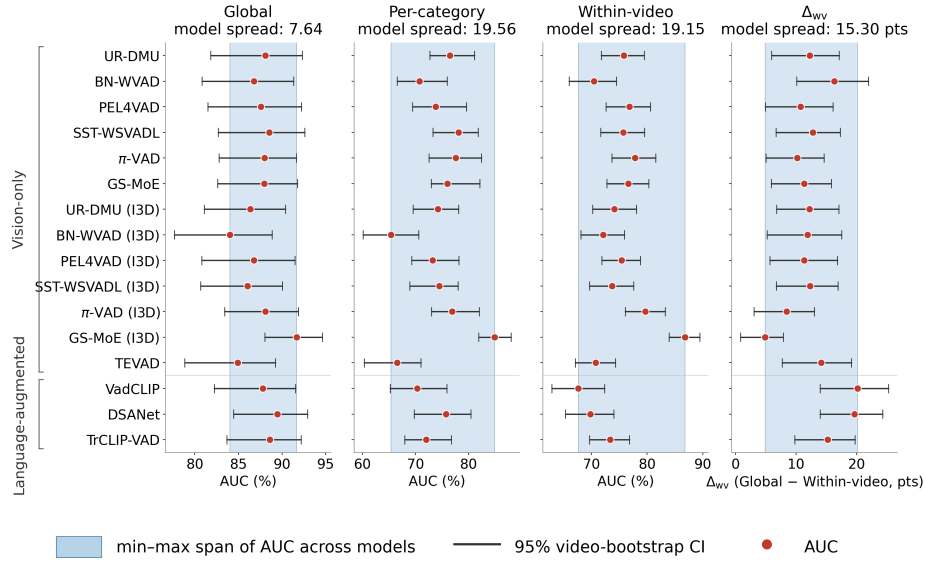


Fig. S1: Score-based AUC spread across the three granularities. The brackets represent the 95% video-bootstrap CI intervals, and the blue band indicates the region where most of these intervals overlap. $\Delta_{wv} = \text{Global} - \text{Within-video}$, is the drop from pooled to same-video comparison; larger values indicate a model whose separation relies more on cross-video pairs than on within-video localization

Why the cross-pole terms cancel. Table S3 shows the four pair-type AUCs, averaged on the top-4 factors with the largest Δ_{scene} gaps. The two cross-pole types move in opposite directions around the same-pole values. In the pooled reference these opposing distortions largely offset, so the net gap stays small even where individual pair types are visibly displaced.

The one factor with a consistent, non-canceling gap is color encoding: Δ_{scene} is positive for every model (mean +0.027, up to +0.069 for TEVAD), meaning the pooled AUC is measurably inflated by color-vs-grayscale comparisons. This validates the audit in Figure 4, where color encoding shows the strongest normal-frame bias. Concurrent activity shows the same effect at smaller magnitude for a subset of models.

Whereas *bias-AUC* establishes that models react to scene factors, Δ_{scene} establishes whether that reaction is load-bearing for detection. On UCF-Crime, under the labeled factors, it is mostly not the case. Nonetheless, color encoding is one measurable exception. After all, it should be emphasized that the dominant confound is not cross-pole but cross-video. Thus, even within one scene pole, the majority of the comparisons still span different recordings. Conditioning on scene factors, therefore, cannot substitute for within-video evaluation.

Table S4: Resolved pairwise global score-AUC comparisons on full 16 models on UCF-Crime test split. All $\binom{16}{2} = 120$ pairs across VideoMAEv2 (6 models), I3D (7 models), and CLIP (3 models) under the paired video bootstrap of Equation (6). **Bold** rows are cross-backbone pairs; plain rows are within-backbone.

Granularity	Winner $\succ$ Loser	Win	Lose	Δ	95% CI
Global	GS-MoE (I3D) $\succ$ BN-WVAD (I3D)	91.65	84.01	+7.88	(+4.77, +11.44)
	GS-MoE (I3D) $\succ$ TEVAD	91.65	84.92	+7.07	(+3.68, +11.13)
	GS-MoE (I3D) $\succ$ SST-WSVADL (I3D)	91.65	86.02	+5.80	(+3.13, +8.45)
	GS-MoE (I3D) $\succ$ UR-DMU (I3D)	91.65	86.35	+5.45	(+2.76, +8.18)
	DSANet $\succ$ BN-WVAD (I3D)	89.44	84.01	+5.45	(+2.29, +8.91)
	GS-MoE (I3D) $\succ$ BN-WVAD	91.65	86.76	+5.05	(+2.24, +8.43)
	GS-MoE (I3D) $\succ$ PEL4VAD (I3D)	91.65	86.76	+4.85	(+2.21, +8.18)
	TrCLIP-VAD $\succ$ BN-WVAD (I3D)	88.59	84.01	+4.65	(+1.35, +8.02)
	DSANet $\succ$ TEVAD	89.44	84.92	+4.63	(+1.29, +8.02)
	SST-WSVADL $\succ$ BN-WVAD (I3D)	88.52	84.01	+4.57	(+1.54, +7.66)
	Pi-VAD (I3D) $\succ$ BN-WVAD (I3D)	88.09	84.01	+4.41	(+0.18, +8.50)
	Pi-VAD $\succ$ BN-WVAD (I3D)	87.99	84.01	+4.18	(+0.58, +7.91)
	GS-MoE (I3D) $\succ$ PEL4VAD	91.65	87.56	+4.18	(+1.25, +7.68)
	GS-MoE (I3D) $\succ$ VadCLIP	91.65	87.79	+4.13	(+1.57, +7.19)
	UR-DMU $\succ$ BN-WVAD (I3D)	88.08	84.01	+4.12	(+0.49, +7.69)
	GS-MoE $\succ$ BN-WVAD (I3D)	87.95	84.01	+4.10	(+0.74, +7.86)
	TrCLIP-VAD $\succ$ TEVAD	88.59	84.92	+3.83	(+0.70, +6.94)
	GS-MoE (I3D) $\succ$ GS-MoE	91.65	87.95	+3.78	(+1.16, +7.08)
	GS-MoE (I3D) $\succ$ UR-DMU	91.65	88.08	+3.76	(+0.64, +7.87)
	VadCLIP $\succ$ BN-WVAD (I3D)	87.79	84.01	+3.76	(+0.89, +6.96)
	SST-WSVADL $\succ$ TEVAD	88.52	84.92	+3.75	(+0.67, +6.94)
	PEL4VAD $\succ$ BN-WVAD (I3D)	87.56	84.01	+3.70	(+0.60, +7.19)
	GS-MoE (I3D) $\succ$ Pi-VAD	91.65	87.99	+3.70	(+1.39, +6.94)
	GS-MoE (I3D) $\succ$ Pi-VAD (I3D)	91.65	88.09	+3.47	(+0.86, +6.60)
	DSANet $\succ$ SST-WSVADL (I3D)	89.44	86.02	+3.37	(+0.45, +6.36)
	GS-MoE (I3D) $\succ$ SST-WSVADL	91.65	88.52	+3.31	(+0.72, +6.62)
	UR-DMU $\succ$ TEVAD	88.08	84.92	+3.31	(+0.01, +6.82)
	GS-MoE (I3D) $\succ$ TrCLIP-VAD	91.65	88.59	+3.23	(+1.09, +5.73)
	PEL4VAD (I3D) $\succ$ BN-WVAD (I3D)	86.76	84.01	+3.04	(+0.42, +5.49)
	VadCLIP $\succ$ TEVAD	87.79	84.92	+2.94	(+0.18, +5.76)
	BN-WVAD $\succ$ BN-WVAD (I3D)	86.76	84.01	+2.83	(+0.04, +5.87)
	DSANet $\succ$ BN-WVAD	89.44	86.76	+2.61	(+0.25, +5.51)
	GS-MoE (I3D) $\succ$ DSANet	91.65	89.44	+2.44	(+0.22, +5.08)
	UR-DMU (I3D) $\succ$ BN-WVAD (I3D)	86.35	84.01	+2.43	(+0.19, +4.89)

Table S5: Resolved pairwise per-category score-AUC comparisons on full 16 models on UCF-Crime test split. All $\binom{16}{2} = 120$ pairs across VideoMAEv2 (6 models), I3D (7 models), and CLIP (3 models) under the paired video bootstrap of Equation (6). **Bold** rows are cross-backbone pairs; plain rows are within-backbone.

Granularity	Winner > Loser	Win	Lose	Δ	95% CI
Granularity	GS-MoE (I3D) > BN-WVAD (I3D)	84.92	65.37	+19.38	(+15.19, +23.78)
	GS-MoE (I3D) > TEVAD	84.92	66.53	+19.26	(+13.56, +24.63)
	GS-MoE (I3D) > VadCLIP	84.92	70.30	+14.50	(+10.34, +19.22)
	GS-MoE (I3D) > BN-WVAD	84.92	70.74	+13.44	(+9.09, +18.06)
	GS-MoE (I3D) > TrCLIP-VAD	84.92	72.01	+12.66	(+8.89, +16.45)
	SST-WSVADL > BN-WVAD (I3D)	78.11	65.37	+12.30	(+6.81, +17.49)
	SST-WSVADL > TEVAD	78.11	66.53	+12.18	(+5.28, +18.05)
	Pi-VAD > BN-WVAD (I3D)	77.58	65.37	+12.14	(+5.98, +18.23)
	Pi-VAD > TEVAD	77.58	66.53	+12.01	(+4.47, +18.83)
	GS-MoE > BN-WVAD (I3D)	75.99	65.37	+11.81	(+5.66, +17.97)
	Pi-VAD (I3D) > BN-WVAD (I3D)	76.88	65.37	+11.80	(+6.24, +17.16)
	GS-MoE > TEVAD	75.99	66.53	+11.69	(+4.60, +18.55)
	Pi-VAD (I3D) > TEVAD	76.88	66.53	+11.68	(+5.47, +17.36)
	UR-DMU > BN-WVAD (I3D)	76.50	65.37	+11.34	(+6.14, +16.91)
	GS-MoE (I3D) > SST-WSVADL (I3D)	84.92	74.46	+11.30	(+7.84, +15.28)
	GS-MoE (I3D) > PEL4VAD (I3D)	84.92	73.22	+11.29	(+7.63, +15.15)
	UR-DMU > TEVAD	76.50	66.53	+11.22	(+4.53, +17.15)
	GS-MoE (I3D) > UR-DMU (I3D)	84.92	74.22	+11.05	(+7.42, +14.89)
	GS-MoE (I3D) > PEL4VAD	84.92	73.78	+10.29	(+5.85, +15.02)
	GS-MoE (I3D) > DSANet	84.92	75.74	+9.71	(+5.41, +14.34)
	DSANet > BN-WVAD (I3D)	75.74	65.37	+9.67	(+3.62, +15.43)
	DSANet > TEVAD	75.74	66.53	+9.55	(+3.32, +15.77)
	PEL4VAD > BN-WVAD (I3D)	73.78	65.37	+9.09	(+3.82, +15.17)
	PEL4VAD > TEVAD	73.78	66.53	+8.96	(+2.04, +15.37)
	UR-DMU (I3D) > BN-WVAD (I3D)	74.22	65.37	+8.33	(+4.54, +12.08)
	UR-DMU (I3D) > TEVAD	74.22	66.53	+8.21	(+2.19, +14.24)
	PEL4VAD (I3D) > BN-WVAD (I3D)	73.22	65.37	+8.09	(+4.17, +12.30)
Per-category	SST-WSVADL (I3D) > BN-WVAD (I3D)	74.46	65.37	+8.08	(+4.01, +12.04)
	GS-MoE (I3D) > UR-DMU	84.92	76.50	+8.04	(+3.97, +12.40)
	PEL4VAD (I3D) > TEVAD	73.22	66.53	+7.97	(+2.09, +13.96)
	SST-WSVADL (I3D) > TEVAD	74.46	66.53	+7.96	(+2.16, +13.36)
	GS-MoE (I3D) > Pi-VAD (I3D)	84.92	76.88	+7.58	(+3.56, +11.68)
	GS-MoE (I3D) > GS-MoE	84.92	75.99	+7.57	(+2.88, +12.02)
	SST-WSVADL > VadCLIP	78.11	70.30	+7.43	(+2.72, +12.25)
	Pi-VAD > VadCLIP	77.58	70.30	+7.26	(+1.61, +12.89)
	GS-MoE (I3D) > Pi-VAD	84.92	77.58	+7.24	(+3.05, +11.89)
	GS-MoE (I3D) > SST-WSVADL	84.92	78.11	+7.08	(+3.33, +11.34)
	GS-MoE > VadCLIP	75.99	70.30	+6.94	(+0.36, +13.23)
	Pi-VAD (I3D) > VadCLIP	76.88	70.30	+6.93	(+1.19, +12.91)
	TrCLIP-VAD > BN-WVAD (I3D)	72.01	65.37	+6.72	(+1.95, +11.68)
	TrCLIP-VAD > TEVAD	72.01	66.53	+6.60	(+0.99, +12.23)
	UR-DMU > VadCLIP	76.50	70.30	+6.47	(+1.09, +11.79)
	SST-WSVADL > BN-WVAD	78.11	70.74	+6.37	(+2.81, +10.05)
	Pi-VAD > BN-WVAD	77.58	70.74	+6.20	(+0.89, +11.81)
	BN-WVAD > BN-WVAD (I3D)	70.74	65.37	+5.94	(+0.59, +11.28)
	GS-MoE > BN-WVAD	75.99	70.74	+5.87	(+1.63, +10.20)
	Pi-VAD (I3D) > BN-WVAD	76.88	70.74	+5.86	(+0.23, +11.48)
	SST-WSVADL > TrCLIP-VAD	78.11	72.01	+5.58	(+1.39, +9.77)
	Pi-VAD > TrCLIP-VAD	77.58	72.01	+5.42	(+0.51, +10.27)
	UR-DMU > BN-WVAD	76.50	70.74	+5.41	(+1.99, +9.12)
	Pi-VAD (I3D) > TrCLIP-VAD	76.88	72.01	+5.08	(+0.17, +10.14)
	DSANet > VadCLIP	75.74	70.30	+4.80	(+1.32, +8.64)
	UR-DMU > TrCLIP-VAD	76.50	72.01	+4.62	(+0.23, +9.13)
	SST-WSVADL > SST-WSVADL (I3D)	78.11	74.46	+4.22	(+0.04, +8.26)
	SST-WSVADL > PEL4VAD	78.11	73.78	+3.22	(+0.43, +6.02)

Table S6: Resolved pairwise within-video score-AUC comparisons on full 16 models on UCF-Crime test split. All $\binom{16}{2} = 120$ pairs across VideoMAEv2 (6 models), I3D (7 models), and CLIP (3 models) under the paired video bootstrap of Equation (6). **Bold** rows are cross-backbone pairs; plain rows are within-backbone.

Granularity	Winner $\succ$ Loser	Win	Lose	Δ	95% CI
Within-video	GS-MoE (I3D) $\succ$ VadCLIP	86.79	67.64	+19.24	(+14.41, +24.03)
	GS-MoE (I3D) $\succ$ DSANet	86.79	69.81	+17.05	(+12.64, +21.40)
	GS-MoE (I3D) $\succ$ BN-WVAD	86.79	70.44	+16.42	(+11.76, +21.25)
	GS-MoE (I3D) $\succ$ TEVAD	86.79	70.79	+16.08	(+12.21, +20.02)
	GS-MoE (I3D) $\succ$ BN-WVAD (I3D)	86.79	72.10	+14.77	(+11.22, +18.34)
	GS-MoE (I3D) $\succ$ TrCLIP-VAD	86.79	73.37	+13.54	(+9.62, +17.43)
	GS-MoE (I3D) $\succ$ SST-WSVADL (I3D)	86.79	73.73	+13.09	(+9.95, +16.63)
	GS-MoE (I3D) $\succ$ UR-DMU (I3D)	86.79	74.13	+12.64	(+9.39, +16.07)
	Pi-VAD (I3D) $\succ$ VadCLIP	79.66	67.64	+12.14	(+7.20, +17.06)
	GS-MoE (I3D) $\succ$ PEL4VAD (I3D)	86.79	75.42	+11.46	(+7.80, +15.16)
	GS-MoE (I3D) $\succ$ SST-WSVADL	86.79	75.75	+11.01	(+7.27, +15.10)
	GS-MoE (I3D) $\succ$ UR-DMU	86.79	75.83	+10.99	(+7.32, +14.90)
	Pi-VAD $\succ$ VadCLIP	77.83	67.64	+10.19	(+4.52, +15.57)
	GS-MoE (I3D) $\succ$ GS-MoE	86.79	76.62	+10.11	(+6.27, +13.88)
	GS-MoE (I3D) $\succ$ PEL4VAD	86.79	76.80	+10.01	(+5.99, +14.12)
	Pi-VAD (I3D) $\succ$ DSANet	79.66	69.81	+9.94	(+4.99, +14.48)
	Pi-VAD (I3D) $\succ$ BN-WVAD	79.66	70.44	+9.31	(+4.09, +14.70)
	PEL4VAD $\succ$ VadCLIP	76.80	67.64	+9.23	(+3.69, +14.55)
	GS-MoE $\succ$ VadCLIP	76.62	67.64	+9.13	(+4.36, +14.07)
	GS-MoE (I3D) $\succ$ Pi-VAD	86.79	77.83	+9.05	(+5.00, +13.18)
	Pi-VAD (I3D) $\succ$ TEVAD	79.66	70.79	+8.97	(+4.52, +13.59)
	UR-DMU $\succ$ VadCLIP	75.83	67.64	+8.25	(+3.83, +12.98)
	SST-WSVADL $\succ$ VadCLIP	75.75	67.64	+8.23	(+3.90, +12.97)
	Pi-VAD $\succ$ DSANet	77.83	69.81	+7.99	(+2.77, +12.93)
	PEL4VAD (I3D) $\succ$ VadCLIP	75.42	67.64	+7.78	(+2.80, +12.81)
	Pi-VAD (I3D) $\succ$ BN-WVAD (I3D)	79.66	72.10	+7.66	(+3.73, +11.51)
	Pi-VAD $\succ$ BN-WVAD	77.83	70.44	+7.36	(+3.24, +11.44)
	GS-MoE (I3D) $\succ$ Pi-VAD (I3D)	86.79	79.66	+7.11	(+4.63, +9.60)
	PEL4VAD $\succ$ DSANet	76.80	69.81	+7.04	(+2.24, +12.00)
	Pi-VAD $\succ$ TEVAD	77.83	70.79	+7.03	(+2.08, +12.06)
	GS-MoE $\succ$ DSANet	76.62	69.81	+6.93	(+2.85, +11.04)
	UR-DMU (I3D) $\succ$ VadCLIP	74.13	67.64	+6.60	(+1.64, +11.81)
	Pi-VAD (I3D) $\succ$ TrCLIP-VAD	79.66	73.37	+6.43	(+2.26, +10.38)
	PEL4VAD $\succ$ BN-WVAD	76.80	70.44	+6.41	(+3.10, +9.70)
	GS-MoE $\succ$ BN-WVAD	76.62	70.44	+6.30	(+2.41, +10.53)
	SST-WSVADL (I3D) $\succ$ VadCLIP	73.73	67.64	+6.15	(+1.17, +11.23)
	PEL4VAD $\succ$ TEVAD	76.80	70.79	+6.07	(+0.96, +11.28)
	UR-DMU $\succ$ DSANet	75.83	69.81	+6.06	(+1.83, +9.93)
	SST-WSVADL $\succ$ DSANet	75.75	69.81	+6.04	(+1.94, +10.01)
	Pi-VAD (I3D) $\succ$ SST-WSVADL (I3D)	79.66	73.73	+5.98	(+2.22, +9.93)
	GS-MoE $\succ$ TEVAD	76.62	70.79	+5.97	(+1.36, +10.90)
	Pi-VAD $\succ$ BN-WVAD (I3D)	77.83	72.10	+5.72	(+1.45, +9.84)
	TrCLIP-VAD $\succ$ VadCLIP	73.37	67.64	+5.71	(+1.63, +9.79)
	PEL4VAD (I3D) $\succ$ DSANet	75.42	69.81	+5.58	(+0.97, +10.12)
	Pi-VAD (I3D) $\succ$ UR-DMU (I3D)	79.66	74.13	+5.54	(+1.64, +9.49)
	UR-DMU $\succ$ BN-WVAD	75.83	70.44	+5.43	(+2.65, +8.33)
	SST-WSVADL $\succ$ BN-WVAD	75.75	70.44	+5.41	(+2.10, +8.74)
	PEL4VAD (I3D) $\succ$ BN-WVAD	75.42	70.44	+4.95	(+0.58, +9.44)
	PEL4VAD $\succ$ BN-WVAD (I3D)	76.80	72.10	+4.76	(+0.56, +8.86)
	GS-MoE $\succ$ BN-WVAD (I3D)	76.62	72.10	+4.66	(+0.35, +9.24)
	PEL4VAD (I3D) $\succ$ TEVAD	75.42	70.79	+4.62	(+0.50, +8.80)
	Pi-VAD (I3D) $\succ$ PEL4VAD (I3D)	79.66	75.42	+4.36	(+0.68, +8.19)
	PEL4VAD (I3D) $\succ$ BN-WVAD (I3D)	75.42	72.10	+3.31	(+0.08, +6.69)

Table S7: Scene pair matrices for all standard UCF scene factors (factor names as in the bias-AUC tables). Shaded rows mark the factor.

Model / factor	w_{AA}	w_{BB}	w_{AB}	w_{BA}	AUC_{AA}	AUC_{BB}	AUC_{AB}	AUC_{BA}	Gap
highway_vs_sidewalk									
UR-DMU [12]	0.372	0.137	0.150	0.341	0.834	0.850	0.831	0.841	-0.004
BN-WVAD [13]	0.372	0.137	0.150	0.341	0.805	0.829	0.799	0.843	0.004
PEL4VAD [6]	0.372	0.137	0.150	0.341	0.833	0.857	0.830	0.850	-0.003
SST-WSVADL [2]	0.372	0.137	0.150	0.341	0.838	0.849	0.837	0.846	-0.001
π -VAD [5]	0.372	0.137	0.150	0.341	0.845	0.837	0.831	0.843	0.000
GS-MoE [1]	0.276	0.225	0.237	0.262	0.913	0.838	0.827	0.918	0.001
STPrompt [9]	0.372	0.137	0.150	0.341	0.877	0.790	0.748	0.908	0.023
VadCLIP [10]	0.373	0.137	0.150	0.341	0.833	0.824	0.787	0.865	0.007
DSANet [11]	0.373	0.137	0.150	0.341	0.862	0.800	0.821	0.849	0.012
TEVAD [3]	0.372	0.137	0.150	0.340	0.802	0.795	0.818	0.790	0.001
TrCLIP-VAD [7]	0.373	0.137	0.150	0.341	0.843	0.851	0.839	0.858	0.002
Universal MIL	0.372	0.137	0.150	0.341	0.859	0.788	0.808	0.842	0.012
closeview_vs_wideview									
UR-DMU [12]	0.207	0.293	0.290	0.209	0.846	0.788	0.847	0.790	0.001
BN-WVAD [13]	0.207	0.293	0.290	0.209	0.834	0.774	0.852	0.762	0.002
PEL4VAD [6]	0.207	0.293	0.290	0.209	0.833	0.792	0.848	0.773	0.000
SST-WSVADL [2]	0.207	0.293	0.290	0.209	0.831	0.778	0.835	0.779	0.001
π -VAD [5]	0.207	0.293	0.290	0.209	0.807	0.765	0.780	0.792	-0.002
GS-MoE [1]	0.227	0.272	0.279	0.222	0.848	0.825	0.852	0.820	0.000
STPrompt [9]	0.207	0.293	0.290	0.209	0.783	0.729	0.821	0.678	0.000
VadCLIP [10]	0.207	0.293	0.290	0.209	0.835	0.734	0.858	0.709	0.001
DSANet [11]	0.207	0.293	0.290	0.209	0.849	0.768	0.835	0.781	-0.002
TEVAD [3]	0.207	0.293	0.290	0.209	0.804	0.736	0.804	0.730	-0.001
TrCLIP-VAD [7]	0.207	0.293	0.290	0.209	0.840	0.757	0.856	0.745	0.002
Universal MIL	0.207	0.293	0.290	0.209	0.804	0.776	0.813	0.769	0.001
singleview_vs_multiview									
UR-DMU [12]	0.475	0.095	0.178	0.253	0.852	0.812	0.793	0.865	0.009
BN-WVAD [13]	0.475	0.095	0.178	0.253	0.833	0.829	0.817	0.839	0.000
PEL4VAD [6]	0.475	0.095	0.178	0.253	0.835	0.833	0.778	0.874	0.001
SST-WSVADL [2]	0.475	0.095	0.178	0.253	0.843	0.812	0.801	0.852	0.007
π -VAD [5]	0.475	0.095	0.178	0.253	0.833	0.770	0.771	0.834	0.015
GS-MoE [1]	0.431	0.112	0.166	0.291	0.829	0.763	0.769	0.823	0.014
STPrompt [9]	0.475	0.095	0.178	0.253	0.783	0.731	0.673	0.820	0.011
VadCLIP [10]	0.475	0.095	0.178	0.253	0.767	0.779	0.683	0.852	0.002
DSANet [11]	0.475	0.095	0.178	0.253	0.823	0.793	0.766	0.843	0.007
TEVAD [3]	0.475	0.095	0.178	0.253	0.792	0.758	0.695	0.834	0.007
TrCLIP-VAD [7]	0.475	0.095	0.178	0.253	0.805	0.800	0.750	0.847	0.003
Universal MIL	0.475	0.095	0.178	0.253	0.838	0.797	0.797	0.836	0.009
color_vs_grayscale									
UR-DMU [12]	0.660	0.033	0.111	0.196	0.809	0.780	0.623	0.898	0.010
BN-WVAD [13]	0.660	0.033	0.111	0.196	0.815	0.733	0.765	0.790	0.028
PEL4VAD [6]	0.660	0.033	0.111	0.196	0.820	0.763	0.763	0.817	0.020
SST-WSVADL [2]	0.660	0.033	0.111	0.196	0.802	0.757	0.668	0.852	0.016
π -VAD [5]	0.660	0.033	0.111	0.196	0.772	0.717	0.630	0.831	0.022
GS-MoE [1]	0.647	0.036	0.114	0.204	0.798	0.740	0.733	0.804	0.021
STPrompt [9]	0.660	0.033	0.111	0.196	0.739	0.669	0.437	0.867	0.024
VadCLIP [10]	0.660	0.033	0.111	0.196	0.785	0.690	0.585	0.846	0.034
DSANet [11]	0.660	0.033	0.111	0.196	0.798	0.734	0.658	0.842	0.023
TEVAD [3]	0.660	0.033	0.111	0.196	0.782	0.593	0.580	0.799	0.069
TrCLIP-VAD [7]	0.660	0.033	0.111	0.196	0.793	0.694	0.647	0.824	0.036
Universal MIL	0.660	0.033	0.111	0.196	0.777	0.717	0.672	0.802	0.021
highresolution_vs_lowresolution									
UR-DMU [12]	0.303	0.194	0.318	0.185	0.794	0.845	0.905	0.689	0.000
BN-WVAD [13]	0.303	0.194	0.318	0.185	0.778	0.818	0.850	0.726	0.001
PEL4VAD [6]	0.303	0.194	0.318	0.185	0.779	0.841	0.879	0.724	0.002
SST-WSVADL [2]	0.303	0.194	0.318	0.185	0.777	0.836	0.887	0.696	0.001
π -VAD [5]	0.303	0.194	0.318	0.185	0.736	0.825	0.837	0.719	0.002
GS-MoE [1]	0.310	0.187	0.325	0.178	0.780	0.868	0.885	0.755	0.002
STPrompt [9]	0.303	0.194	0.318	0.185	0.704	0.782	0.882	0.538	0.002
VadCLIP [10]	0.303	0.194	0.318	0.185	0.742	0.807	0.877	0.625	0.002
DSANet [11]	0.303	0.194	0.318	0.185	0.761	0.823	0.866	0.700	0.003
TEVAD [3]	0.303	0.194	0.318	0.185	0.721	0.791	0.856	0.630	0.004
TrCLIP-VAD [7]	0.303	0.194	0.318	0.185	0.755	0.817	0.871	0.671	0.003
Universal MIL	0.303	0.194	0.318	0.185	0.744	0.814	0.868	0.661	0.002
outdoor_vs_indoor									
UR-DMU [12]	0.199	0.301	0.199	0.301	0.871	0.788	0.823	0.840	-0.002
BN-WVAD [13]	0.199	0.301	0.199	0.301	0.854	0.758	0.732	0.869	-0.001
PEL4VAD [6]	0.199	0.301	0.199	0.301	0.872	0.776	0.801	0.848	-0.002
SST-WSVADL [2]	0.199	0.301	0.199	0.301	0.858	0.786	0.813	0.831	-0.003
π -VAD [5]	0.199	0.301	0.199	0.301	0.831	0.758	0.785	0.808	-0.002
GS-MoE [1]	0.210	0.293	0.236	0.261	0.814	0.825	0.795	0.842	0.001
STPrompt [9]	0.199	0.301	0.199	0.301	0.794	0.722	0.742	0.776	-0.001
VadCLIP [10]	0.199	0.301	0.199	0.301	0.814	0.729	0.751	0.785	-0.004
DSANet [11]	0.199	0.301	0.199	0.301	0.838	0.767	0.793	0.814	-0.002
TEVAD [3]	0.199	0.301	0.199	0.301	0.825	0.706	0.755	0.784	-0.003
TrCLIP-VAD [7]	0.199	0.301	0.199	0.301	0.833	0.755	0.774	0.815	-0.002
Universal MIL	0.199	0.301	0.199	0.301	0.838	0.767	0.768	0.837	0.000

Table S8: Table S7 continued. Scene pair matrices for all standard UCF scene factors (factor names as in the bias-AUC tables). Shaded rows mark the factor.

Model / factor	w_{AA}	w_{BB}	w_{AB}	w_{BA}	AUC_{AA}	AUC_{BB}	AUC_{AB}	AUC_{BA}	Gap
few-person_vs_crowd									
UR-DMU [12]	0.441	0.103	0.307	0.149	0.778	0.864	0.850	0.798	-0.009
BN-WVAD [13]	0.441	0.103	0.307	0.149	0.767	0.847	0.819	0.805	-0.010
PEL4VAD [6]	0.441	0.103	0.307	0.149	0.780	0.849	0.803	0.840	-0.011
SST-WSVADL [2]	0.441	0.103	0.307	0.149	0.769	0.853	0.824	0.809	-0.010
π -VAD [5]	0.441	0.103	0.307	0.149	0.742	0.833	0.812	0.768	-0.011
GS-MoE [1]	0.454	0.103	0.272	0.171	0.825	0.851	0.844	0.833	-0.004
STPrompt [9]	0.441	0.103	0.307	0.149	0.699	0.820	0.774	0.776	-0.014
VadCLIP [10]	0.441	0.103	0.307	0.149	0.745	0.836	0.800	0.790	-0.013
DSANet [11]	0.441	0.103	0.307	0.149	0.775	0.856	0.798	0.853	-0.014
TEVAD [3]	0.441	0.103	0.307	0.149	0.708	0.835	0.800	0.761	-0.014
TrCLIP-VAD [7]	0.441	0.103	0.307	0.149	0.763	0.847	0.812	0.812	-0.011
Universal MIL	0.441	0.103	0.307	0.149	0.753	0.831	0.793	0.806	-0.011
singleact_vs_multiact									
UR-DMU [12]	0.223	0.239	0.408	0.131	0.759	0.797	0.836	0.707	0.015
BN-WVAD [13]	0.223	0.239	0.408	0.131	0.768	0.767	0.800	0.721	0.007
PEL4VAD [6]	0.223	0.239	0.408	0.131	0.786	0.781	0.787	0.770	-0.000
SST-WSVADL [2]	0.223	0.239	0.408	0.131	0.764	0.782	0.794	0.747	0.005
π -VAD [5]	0.223	0.239	0.408	0.131	0.716	0.768	0.765	0.719	0.007
GS-MoE [1]	0.258	0.205	0.408	0.129	0.785	0.823	0.822	0.788	0.004
STPrompt [9]	0.223	0.239	0.408	0.131	0.684	0.746	0.780	0.636	0.017
VadCLIP [10]	0.223	0.239	0.408	0.130	0.731	0.724	0.827	0.610	0.025
DSANet [11]	0.223	0.239	0.408	0.130	0.771	0.771	0.797	0.731	0.005
TEVAD [3]	0.223	0.239	0.408	0.130	0.686	0.741	0.809	0.606	0.026
TrCLIP-VAD [7]	0.223	0.239	0.408	0.130	0.748	0.762	0.814	0.678	0.014
Universal MIL	0.223	0.239	0.408	0.131	0.731	0.783	0.777	0.737	0.006
highmotion_vs_lowmotion									
UR-DMU [12]	0.329	0.181	0.221	0.269	0.822	0.818	0.768	0.861	0.000
BN-WVAD [13]	0.329	0.181	0.221	0.269	0.809	0.792	0.704	0.869	-0.001
PEL4VAD [6]	0.329	0.181	0.221	0.269	0.818	0.809	0.726	0.879	-0.001
SST-WSVADL [2]	0.329	0.181	0.221	0.269	0.813	0.804	0.738	0.864	-0.000
π -VAD [5]	0.329	0.181	0.221	0.269	0.803	0.753	0.721	0.826	0.004
GS-MoE [1]	0.301	0.202	0.217	0.280	0.840	0.808	0.754	0.880	0.002
STPrompt [9]	0.329	0.181	0.221	0.269	0.762	0.722	0.673	0.799	0.003
VadCLIP [10]	0.329	0.181	0.221	0.269	0.788	0.761	0.701	0.832	0.002
DSANet [11]	0.329	0.181	0.221	0.269	0.815	0.767	0.721	0.849	0.004
TEVAD [3]	0.329	0.181	0.221	0.269	0.772	0.746	0.694	0.811	0.002
TrCLIP-VAD [7]	0.329	0.181	0.221	0.269	0.806	0.764	0.726	0.835	0.003
Universal MIL	0.329	0.181	0.221	0.269	0.790	0.771	0.702	0.843	0.001
shopshelves_vs_cashier									
UR-DMU [12]	0.330	0.181	0.244	0.245	0.705	0.850	0.731	0.834	-0.008
BN-WVAD [13]	0.330	0.181	0.244	0.245	0.754	0.839	0.674	0.898	-0.011
PEL4VAD [6]	0.330	0.181	0.244	0.245	0.709	0.853	0.740	0.820	-0.011
SST-WSVADL [2]	0.330	0.181	0.244	0.245	0.697	0.855	0.781	0.757	-0.015
π -VAD [5]	0.330	0.181	0.244	0.245	0.672	0.811	0.728	0.764	-0.008
GS-MoE [1]	0.227	0.271	0.209	0.293	0.830	0.857	0.807	0.873	0.002
STPrompt [9]	0.330	0.181	0.244	0.245	0.738	0.795	0.740	0.797	-0.003
VadCLIP [10]	0.330	0.181	0.244	0.245	0.796	0.841	0.795	0.840	-0.004
DSANet [11]	0.330	0.181	0.244	0.245	0.811	0.850	0.809	0.847	-0.004
TEVAD [3]	0.330	0.181	0.244	0.245	0.750	0.771	0.723	0.798	-0.001
TrCLIP-VAD [7]	0.330	0.181	0.244	0.245	0.809	0.837	0.772	0.861	-0.005
Universal MIL	0.330	0.181	0.244	0.245	0.690	0.824	0.694	0.821	-0.010
residential_area_vs_gasstation									
UR-DMU [12]	0.805	0.010	0.107	0.078	0.771	0.864	0.894	0.699	-0.038
BN-WVAD [13]	0.805	0.010	0.107	0.078	0.730	0.880	0.614	0.963	-0.067
PEL4VAD [6]	0.805	0.010	0.107	0.078	0.780	0.869	0.750	0.921	-0.036
SST-WSVADL [2]	0.805	0.010	0.107	0.078	0.736	0.851	0.860	0.706	-0.046
π -VAD [5]	0.805	0.010	0.107	0.078	0.695	0.800	0.812	0.687	-0.040
GS-MoE [1]	0.707	0.022	0.185	0.085	0.812	0.840	0.830	0.835	-0.008
STPrompt [9]	0.805	0.010	0.107	0.078	0.664	0.757	0.744	0.673	-0.036
VadCLIP [10]	0.805	0.010	0.107	0.078	0.722	0.798	0.788	0.735	-0.029
DSANet [11]	0.805	0.010	0.107	0.078	0.734	0.807	0.797	0.748	-0.028
TEVAD [3]	0.805	0.010	0.107	0.078	0.611	0.794	0.861	0.549	-0.068
TrCLIP-VAD [7]	0.805	0.010	0.107	0.078	0.681	0.840	0.811	0.749	-0.059
Universal MIL	0.805	0.010	0.107	0.078	0.742	0.809	0.782	0.772	-0.026
residential_area_vs_commercial									
UR-DMU [12]	0.242	0.257	0.272	0.229	0.771	0.705	0.903	0.499	-0.010
BN-WVAD [13]	0.242	0.257	0.272	0.229	0.730	0.753	0.699	0.794	0.001
PEL4VAD [6]	0.242	0.257	0.272	0.229	0.780	0.709	0.768	0.699	-0.004
SST-WSVADL [2]	0.242	0.257	0.272	0.229	0.736	0.697	0.774	0.604	-0.010
π -VAD [5]	0.242	0.257	0.272	0.229	0.695	0.671	0.755	0.609	0.002
GS-MoE [1]	0.264	0.232	0.299	0.204	0.812	0.830	0.842	0.796	0.001
STPrompt [9]	0.242	0.257	0.272	0.229	0.664	0.738	0.886	0.422	-0.013
VadCLIP [10]	0.242	0.257	0.272	0.229	0.722	0.795	0.857	0.633	-0.001
DSANet [11]	0.242	0.257	0.272	0.229	0.734	0.810	0.821	0.725	0.003
TEVAD [3]	0.242	0.257	0.272	0.229	0.611	0.750	0.816	0.524	0.002
TrCLIP-VAD [7]	0.242	0.257	0.272	0.229	0.681	0.808	0.837	0.656	0.006
Universal MIL	0.242	0.257	0.272	0.229	0.742	0.690	0.799	0.578	-0.009

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